

Data Table Documentation

Introduction

This document provides an overview of the data tables used in this project. The data tables are organized into two folders: `kernel` and `extended`. You can find all the hydraulic objects in the map `Tokke_topopoly.pdf` and the physical configurations in `Tokke_Vinje_topology.yaml`. Both are in the `extended` folder. The dataset is generated based on Tokke-Vinje watercourse v5_2026_6 by SHOP v17.17.0.

Market price

File: `/data/kernel/Historical_day_ahead_price_2015_2025.csv`

date

UTC timestamp for each hour.

Day-ahead price (EUR/MWh)

Day-ahead market price for selling electricity at each hour. Unit: EUR/MWh.

Reservoir and river inflow

File: `/data/kernel/Historical_inflow_1958_2025.csv`

date

UTC timestamp for each hour.

Reservoir names (e.g., Langesae)

Inflow into each reservoir at each hour. Unit: m³/s. There are 17 reservoirs.

r_Vest_Vassdraget

Inflow to the river at each hour. Unit: m³/s. Only one river has inflow.

Reservoir storage

File: `/data/kernel/Historical_volume_2015_2024.csv`

date

UTC timestamp for each day.

Reservoir names (e.g., Kjelavatn)

Water storage for each reservoir at the start of the day. Units: Million m³. There are 17 reservoirs. Note that because there are no historical records for two small reservoirs, Hyljelihyl and Vatjern, their storage is assumed to be 50% full.

Reservoir water value

File: /data/kernel/Synthetic_water_value_2015_2024.csv

date

UTC timestamp for each day.

Reservoir names (e.g., Kjelavatn)

Marginal water value for each reservoir at the start of the day. Unit: EUR/MWh. There are 17 reservoirs. For a scheduling problem spanning one week, for example, starting at 2015-01-01 00:00 and ending at 2015-01-07 23:00, the initial volume of a reservoir is the value at 2015-01-01 00:00, and its final water value should be the value at 2015-01-08 00:00.

Reservoir minimum volume

File: /data/kernel/Constraint_min_volume.csv

date

UTC timestamp for each hour.

Reservoir names (e.g., Totak)

Minimum water storage for each reservoir at each hour. Unit: Million m³. Only 3 reservoirs have minimum storage requirements in some periods of the year. If there are no restrictions, the value is 0.

River minimum flow

File: /data/kernel/Constraint_min_flow.csv

date

UTC timestamp for each hour.

River names (e.g., b_Kjelavatn)

Minimum flow requirement for each river at each hour. Unit: m³/s. Six rivers have minimum flow requirements in some periods of the year. If there are no restrictions, the value is 0.

Unit commitment decisions

File: /data/kernel/Unit_commitment_decisions.csv

Run No

Unique identifier for each case.

starttime

The start time (UTC) for each case.

total_calculation_time

The total time is used to solve this case. Unit: second.

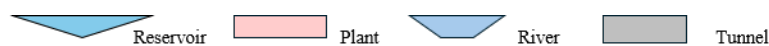
result_committed_XX_YY_ZZ

Unit commitment decisions for each unit at each hour. A value of 0 or 1 indicates “off” or “on”. XX is the name of the hydropower plant. YY is the generating unit. ZZ is the hour. For example, result_committed_Hogga_G1_t0 refers to the unit commitment decision for Plant Hogga, Unit G1, at the first hour.

Tokke-Vinje watercourse topology

File: /data/extended/Tokke_topology.pdf

Visualize the Tokke-Vinje watercourse, where



Tokke-Vinje watercourse physical configurations

File: /data/extended/Tokke_Vinje_topology.yaml

This YAML file is provided in a SHOP-ready format. It contains all the information required for general case-study use. Definitions of all topology attributes are available in the SHOP documentation: <https://docs.shop.sintef.energy/intro.html>